Dylan Spektor

spektordylan@gmail.com | 720-498-5441 | Boulder, CO | LinkedIn | GitHub | Website

EDUCATION & HONORS

University of Colorado Boulder **Expected Apr 2028**
B.S. Applied Mathematics & B.A. Computer Science
GPA: 4.0
Organizations: CU Quants, Engineering Honors Program, Tau Beta Pi, Quantum Scholars.
Relevant Coursework: Mathematical Finance, Probability & Statistics, Algorithms, Neural Networks & Deep Learning, Numerical Methods, Applied Real Analysis, Advanced Data Science. Upcoming independent study in Machine Learning in Finance.

Colorado Early Colleges Douglas County North **May 2025**
A.S. Computer Science (Dual Enrollment at Arapahoe Community College) | 1580 SAT, National Merit Finalist

RESEARCH & QUANTITATIVE EXPERIENCE

CU Quants Fund **Oct 2025 – Present**
Quantitative Developer
- Researching and developing volatility forecasting methods for cryptocurrency market making, with applications to perpetual-futures hedging. (Currently reading these papers).
- Design and implement data pipelines over historical order-book data to support research into stablecoin spread modeling and market-making opportunities.
- Tested crypto lead-lag predictive signals using cross-correlation and transaction costs; prototyped convex capital allocation across pairs strategies.

Undergraduate Researcher — Programming Languages & Verification Group **Aug 2026 – Apr 2027**
Advised by Prof. Evan Chang, CU Boulder
- Selected for a yearlong research apprenticeship contributing to Demanded Abstract Interpretation of higher-order languages; targeting second authorship on a planned November 2026 PLDI submission.
- Exploring formal verification of AI agents under specification ambiguity.

SELECTED PROJECTS & RESEARCH

Treasury Portfolio Optimization Calculator **Oct 2026**
- Built a minimum-cost Treasury dedication optimizer matching liability cash flows with mixed-integer linear programming, integer bond lots, and rolling surplus cash.
- Implemented Treasury cash-flow schedules and accrued-interest/dirty-price calculations with QuantLib; CLI ingests market data and outputs optimal CUSIP/principal allocations.

Spectral Bias in Neural PDE Solvers **Current**
- Investigating spectral bias in neural networks for the heat and wave equations through Fourier/Sturm–Liouville analysis and its relationship to Neural Tangent Kernel (NTK) theory.
- Extending the analysis to Physics-Informed Neural Networks (PINNs), studying spectral bias in PDE residuals and its relationship to solution error, with computational demonstrations of mitigation methods planned.

SMT-Based Cyclic Arbitrage Detection **Jul 2026**
- Built a Z3 SMT-based cross-currency arbitrage detector supporting fixed fees, nonlinear slippage, and integer lot sizes; validated exact agreement with Bellman-Ford on a frictionless baseline, then identified a capital-dependent profitability window confirmed by brute-force enumeration.

Riemann Zeta Zeros & Random-Matrix Spectral Statistics **Apr 2026**
- Co-authored a complex-analysis report relating Riemann zeta zeros to GUE/CUE spectral statistics in random matrix theory and the Hilbert–Pólya conjecture; derived moment-based connections and outlined a contour-integration approach to zero-spacing.

External Quantum Efficiency Degradation in Perovskite Cells **Dec 2025**
- In an experimental collaboration with the National Laboratory of the Rockies, analyzed weekly 350–400 nm EQE measurements across 0, 5, and 10 deposition cycles using bootstrap resampling, linear/quadratic regression, and ANOVA; found statistically significant differences in degradation coefficients.

TECHNICAL SKILLS

Programming & Data: Python (NumPy, pandas, SciPy, scikit-learn, statsmodels, QuantLib), Lean 4, Z3/SMT, functional programming
Statistics & ML: probability, statistics, machine learning, deep learning, predictive modeling, time-series analysis, regression, bootstrapping, hypothesis testing, cross-correlation
Optimization & Algorithms: linear/mixed-integer programming, convex optimization, graph algorithms, constraint solving
Research Methods: Fourier/spectral analysis, random-matrix theory, formal verification, abstract interpretation